AegisFlow-AI: AI-Based Real-Time Intrusion Detection System

Technical Proposal & Architectural Blueprint

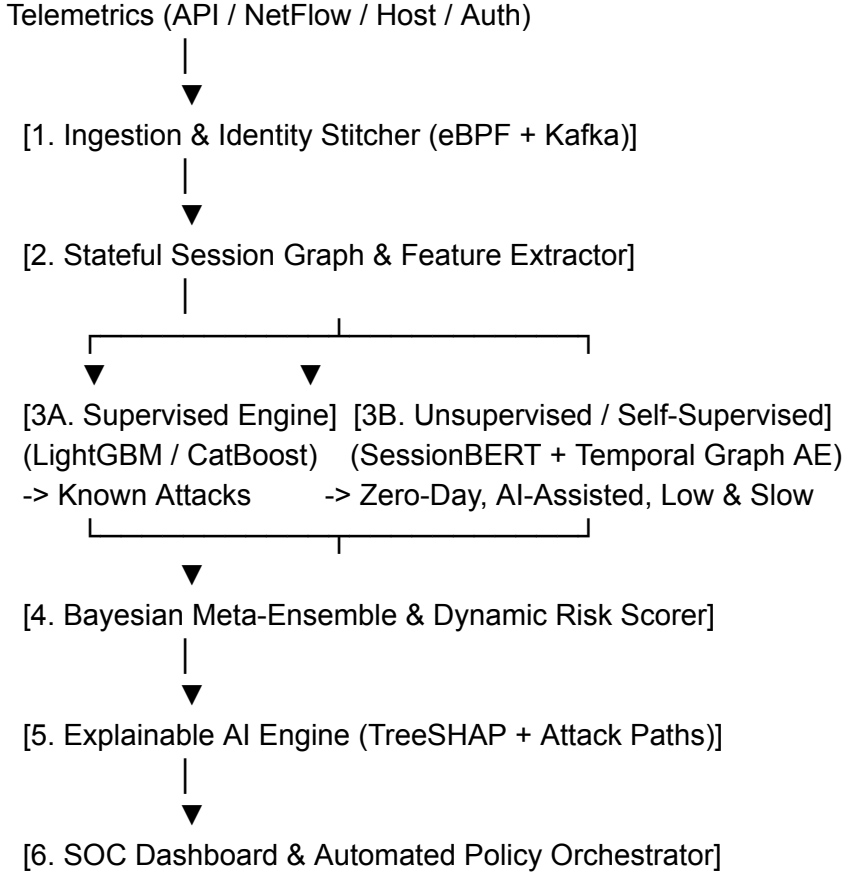
1. Executive Summary & Core Innovation

Modern cyber-attacks have evolved into multi-stage, low-and-slow, and AI-assisted campaigns that easily evade traditional signature-based Intrusion Detection Systems (IDS). These threats often appear completely benign when individual packets, requests, or events are inspected in isolation.

AegisFlow-AI is a real-time behavioral intrusion detection and incident correlation platform. Instead of stateless packet-level inspection, it introduces a **Dual-Core Behavioral AI Engine**:

1. **Dynamic Temporal Graph Convolutional Network (T-GCN):** Tracks relational entity mapping across users, tokens, IP addresses, and endpoints over time.
2. **Self-Supervised Sequence Engine (SessionBERT):** Treats user session trails and API navigation flows like language tokens to capture semantic anomalies, multi-stage kill chains, and low-and-slow zero-day exploits.

2. End-to-End System Architecture



3. Core Modules & Methodology

Module 1: Session Reconstruction & Event Correlation

- **6-Tuple Context Stitching:** Correlates disparate network, host, and application events using (UserID, AuthToken/JWT, Device-Fingerprint, Src-IP, Dst-IP, User-Agent) into continuous, stateful session graphs.
- **Low-Overhead Telemetry Capture:** Utilizes **eBPF** in kernel space to observe socket states, system calls, and network flows with near-zero latency overhead.
- **Stream Synchronization:** Employs **Apache Kafka & Apache Flink** with sliding time windows (50ms–100ms) to reconstruct chronologically ordered session streams.

Module 2: Behavioral Feature Extraction & Selection

- **Flow & Protocol Telemetry:** Packet size entropy, Inter-Arrival Time (IAT), Bytes in/out ratios, connection duration, and TCP window characteristics.
- **Temporal & Statistical Rates:** Burstiness scores, destination IP diversity, unique port-access frequencies, and connection failure ratios across sliding temporal buckets (5m, 15m, 60m).
- **Application & API Navigation Metrics:**
    - *Endpoint Transition Likelihood:* Markov-based regular API flow transition vs. abrupt privilege escalation jumps.
    - *Object-ID Enumeration Index:* Velocity and pattern of unique resource access requests.
    - *Token Disparity:* Geographic/IP divergence on active authorization tokens.
- **Feature Importance Analysis:** Utilizes Mutual Information (MI) and Recursive Feature Elimination (RFE) to isolate the most discriminative attack features in real time.

Module 3: Dual-Core AI Behavioral Detection Engine

- **Known Attack Classification:** High-speed gradient boosting models (LightGBM/CatBoost) trained on benchmark data (CIC-IDS2018, UNSW-NB15) to detect recognized DDoS, Brute-Force, and Botnet patterns in under 5ms.
- **Multi-Step & Low-and-Slow Attack Engine (SessionBERT):** A transformer model trained on legitimate API execution sequences. If an adversary carries out stealthy reconnaissance spread across 30 minutes, the sequence likelihood score collapses, triggering an anomaly flag.
- **Emerging & AI-Assisted Attack Detection (Graph Autoencoder):** Maps cross-layer telemetry into dynamic sub-graphs. AI-driven attack vectors using randomized jitter or polymorphic payloads yield massive reconstruction errors within the graph latent space.

Module 4: Dynamic Threat Scoring & Risk Prioritization

- **Bayesian Fusion Scoring:** Computes an aggregated session-level risk score normalized from 0 to 100:

- $RiskScore = w_1 \cdot P_{Known} + w_2 \cdot Anomaly_{Seq} + w_3 \cdot Anomaly_{Graph}$
- **Contextual Severity Escalation:** Sessions accessing crown-jewel assets (e.g., /admin/export, /iam/credentials) trigger instant multiplier elevation for critical prioritization.

Module 5: Explainable AI (XAI) & Incident Reconstruction

- **Attack Path Traversal:** Displays the reconstructed timeline for SOC analysts: Failed Logins (3x) -> Successful Login -> Token Divergence -> Rapid IDOR Access -> Large Egress
- **Attribution Breakdown:** TreeSHAP and Path Attribution show exact percentage contributions of

each feature to the alert, eliminating black-box opacity.

4. Benchmark & Unseen-Attack Evaluation Plan

Experiment Phase	Training Dataset	Testing Dataset	Target Benchmark
Phase 1: Known Attack Classification	CIC-IDS2018 / UNSW-NB15 (80% Normal + Known Attacks)	20% Unseen slices of Known Attacks	Accuracy > 99%, F1-Score > 0.98
Phase 2: Zero-Day & Unseen Attack Test	Baseline Normal Traffic + Basic DoS/DDoS (All advanced vectors excluded)	Baseline Normal + **Infiltration / Botnet / IDOR sequences** (Novel Threats)	Anomaly Recall > 93%, False Positive Rate (FPR) < 1.5%
Phase 3: Low-and-Slow Temporal Resilience	Normal User Session Trails	Multi-stage attacks spread across 15–45 min intervals	Mean Time to Detect (MTTD) < 3 seconds on final component

5. Technology Stack

- **Ingestion & Streaming:** eBPF, Zeek, Fluentbit, Apache Kafka, Apache Flink
- **Machine Learning & AI:** PyTorch (T-GCN, SessionBERT), Hugging Face, LightGBM, SHAP, NetworkX
- **Data Storage:** ClickHouse (Time-series telemetry), Neo4j (Session graphs), Redis (Stateful caching)
- **API & Frontend:** FastAPI, React, D3.js (Interactive graph visualizer & timeline), WebSockets

6. Key Differentiators (Pitch Highlights)

1. **Journey-Centric, Not Packet-Centric:** Correlates disparate network and API actions across user sessions rather than evaluating stateless packets.
2. **Signature-Free Zero-Day Defense:** SessionBERT and Graph Autoencoders identify novel attack variations and AI-assisted automation without relying on pre-existing signatures.
3. **Transparent & SOC-Ready (XAI):** Provides visual attack trees with mathematical feature attribution, drastically reducing analyst triage time.
4. **Sub-Second Real-Time Telemetry:** Optimized with eBPF and streaming pipelines to deliver near-instantaneous anomaly detection at line rate.